

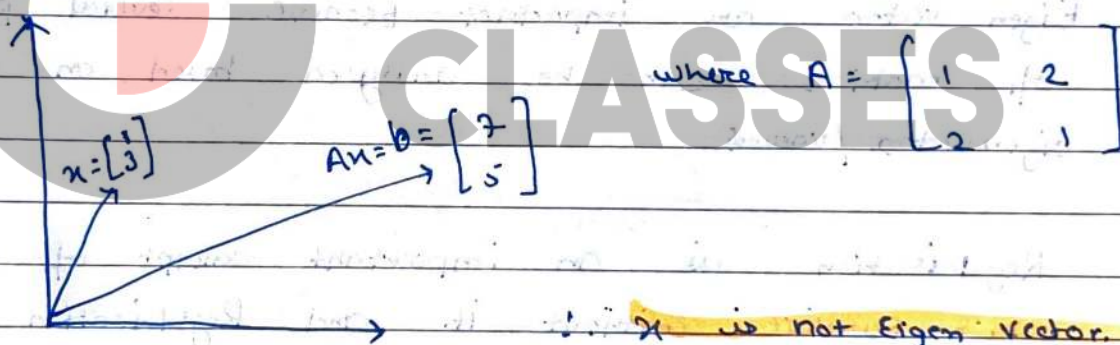
Part 4Eigen Values & Eigen Vectors

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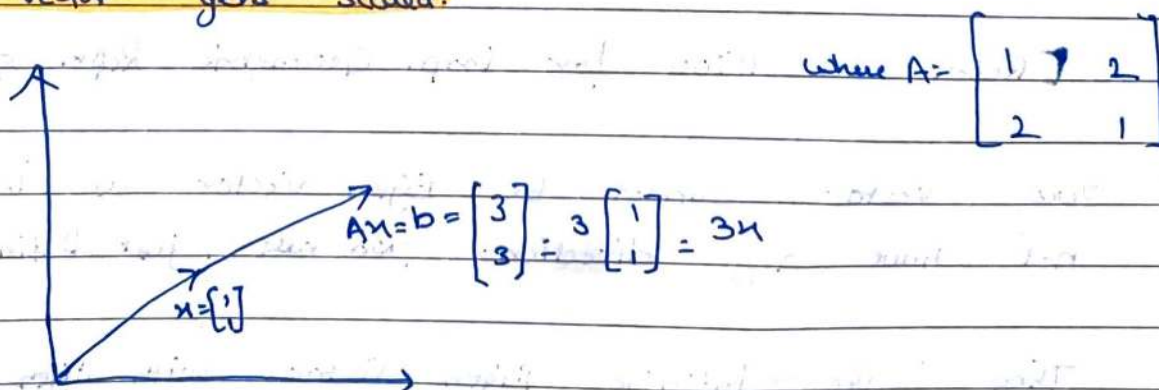
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- Rank was one of the way to look at a Matrix & Eigen Values are new way to look at the same matrix.
- All matrices dealt in this chapter are square matrices.
- What happens when a matrix hits a vector? Generally, we know it makes L.C. of matrix. But Geometric pov have two probabilities:

P1: Vectors gets Transformed into a new vector:



P2: Vector gets Scaled.



∴ x is EV.

So here in P2 x is called Eigen Vector for A .

So vectors which have the same direction after multiplication with matrix are called Eigen Vectors.

So if $Ax = \lambda x$, where λ is scalar, then x is Eigen vector of A & λ is Eigen Value. λ can be zero also.

Now if $\lambda = 1$, then no change in x .

if $\lambda > 1$, then x gets stretched.

if $\lambda < 1$, then x gets shrinked.

→ Eigen vectors are important because several properties of matrices can be analysed based on their eigen vector / values.

→ Regularisation is an important concept of ML. ML can't survive without it. And Regularisation is completely based on Eigen Values & Vectors. Many other important applications of EVs are there.

[Refer SA 15:00 for imp. Geometric Repr. of EV].

* Zero vector cannot be Eigen Vector as it does not have any direction. (No rule, just Definition)

* There are infinite Eigen Vectors with Eigen Value = 1 for identity matrices. But for any matrix if there is at least one EV then there it also have infinitely many. (Either 0 or infinite).

EV - Eigen Vector E Val - Eigen Value

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NOTE
↔

Eigen vectors only makes sense for square Matrices

$$\underset{m \times m}{A} \underset{n \times 1}{x} = \underset{m \times 1}{\lambda x'}$$

But these two are not same bcz they are in two different spaces.

Continuing Last Page's Last.

Every vector is Eigen Vector for Identity Matrix.

Also, Every matrix have infinite but not each vector is Eigen Vector in that case.

And i.e. infinite bcz if we find one then we can take any l.c. of that and that would also be EV. i.e. if x is EV then kx is also EV
Vector scalar

In Identity Matrix of size $n \times n$, n L.D. vectors can be there. So, we are not interested in no. of EV, we are interested in no. of L.D. EVs, because L.D. EVs can be infinite.

★

Calculating Eigen values & Eigen Vectors:

$$Ax = \lambda x \quad (\text{Def. of EV})$$

$$Ax - \lambda x = 0$$

$$(A - \lambda I)x = 0 \quad - (1) \quad (\text{I bcz Matrix Subtraction})$$

and $x \neq 0$ bcz x is Eigen Vector & eigen vector can't be zero.

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So, $\lambda \neq 0$ then, $|A - \lambda I| = 0 \rightarrow$ Characteristic EquationSo, we will solve char Eq. let's call it Matrix B, and get λ .then $Bx = 0$ (put λ in B now) (eq-1) $\rightarrow$ Just Homogeneous S.O.E, so we will get λ which is EV.See this instead of def.Ex: find Eigenvalues & Eigenvectors for the following matrix:

a)

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

Step 1. Solve $|A - \lambda I| = 0$

$$\Rightarrow \left| \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)^2 - 4 = 0$$

$$\Rightarrow (1-\lambda)^2 = 4$$

$$\Rightarrow 1-\lambda = \pm 2$$

$$\Rightarrow 1-\lambda = \pm 2$$

$$\underline{\lambda = -1}, \underline{\lambda = 3}$$

Step 2 Now find matrix B i.e. $A - dI$ by putting d

for $d = -1$

$$\text{So, } B = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$Bx = 0 \Rightarrow \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\hookrightarrow$ ~~AB~~

Step 3 Now solve above S.O.E. (As we need E.V. i.e. x)
as we have $Bx = 0$

$$\text{Aug. Matrix } \begin{bmatrix} 2 & 2 & | & 0 \\ 2 & 2 & | & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 2 & 2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\text{Now } x_2 = k$$

$$\text{or } 2x_1 + 2x_2 = 0$$

$$2x_1 = -2k$$

$$x_1 = -k$$

So

$$x = k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Now for $d = -1$, EV = $k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, i.e. one L.E. EV.

$\hookrightarrow$ Always be present.

bec free var. are always present

$\hookrightarrow$ as det. is zero

Similarly we can calculate EV for $d = 3$.

$$\text{it will be } x = k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

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b) $A = \begin{bmatrix} 9 & 0 & -8 \\ 15 & 3 & -15 \\ 0 & 0 & 1 \end{bmatrix}$

Step-1. $|A - \lambda I| = 0$

$$\begin{vmatrix} 9-\lambda & 0 & -8 \\ 15 & 3-\lambda & -15 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$(9-\lambda)[(3-\lambda)(1-\lambda)] - 8[0] = 0$$

$$(9-\lambda)(3-\lambda)(1-\lambda) = 0$$

So $\lambda = 1, 3, 9$

Step 2 Find B

for $\lambda = 1$

$$\begin{bmatrix} 8 & 0 & -8 \\ 15 & 2 & -15 \\ 0 & 0 & 0 \end{bmatrix}$$

$B\eta = 0$

$$\left[\begin{array}{ccc|c} 8 & 0 & -8 & 0 \\ 15 & 2 & -15 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - \frac{15}{8}R_1$$

$$\left[\begin{array}{ccc|c} 8 & 0 & -8 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

So $x_3 = k$, $8x_1 - 8x_2 = 0$

$2x_2 = 0$

$8x_1 = 8x_2$

$x_2 = 0$

$x_1 = x_3$

So $\lambda = \begin{bmatrix} 1 \\ k \\ 0 \\ 1 \end{bmatrix}$

EVI

Similarly find for $\lambda = 3$ & $\lambda = 9$

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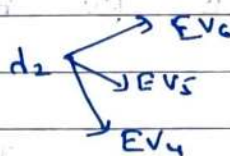
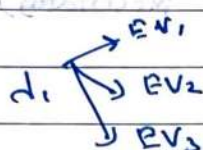
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Imp

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Eigen vectors from different ~~val~~ eigenvalues are linearly independent.

i.e. if $d_1 \neq d_2$



(Optional)

then EV_1, EV_2 & EV_3 are LD but EV_4 & EV_5 are LI.

Proof:

given: $d_1 \neq d_2$

to prove: e_1 to e_2 are LI

For Contradiction: let $e_1 = k e_2$ (assume)

then $A e_1 = A (k e_2) = k A e_2$ - (i)

$A e_1 = d_1 e_1$

Also $A e_1 = A (k e_2) = k A e_2 = k d_2 e_2$ - (ii)

From (i) & (ii)

$$d_1 k e_2 = d_2 k e_2$$

$$d_1 e_2 - d_2 e_2 = 0$$

$$(d_1 - d_2) e_2 = 0$$

Since e_2 is EV, it can't be zero

$$d_1 - d_2 = 0$$

$$d_1 = d_2$$

So, eigen values are same in this case.

but given $d_1 \neq d_2$, so assumption failed.

$\therefore$ Hence Proved.

(Can also be proved by examples)

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Ques:

* If λ is repeating, then we may get more than one LI EV too. (Not Necessary) (Atleast 1)

Ex:

$$A = \begin{bmatrix} 5 & -1 & 0 \\ -1 & 5 & 0 \\ 1/3 & -1/3 & 4 \end{bmatrix}$$

on finding $|A - \lambda I| = 0$

we got

$$(\lambda - 6)(\lambda - 4)^2 = 0$$

So

$$\lambda = 6, 4, 4$$

$$\lambda_1 = 6, \lambda_2 = 4, \lambda_3 = 4$$

for

$$\lambda = 6$$

EV =

$$K \begin{bmatrix} 3 \\ -3 \\ 1 \end{bmatrix}$$

&

$$\text{for } \lambda = 4$$

EV =

$$s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

So two LI vector here in EV. $\lambda = 4$, because we had λ_1, λ_2 here. So it can be possible.

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Ques: $(\lambda-1)^2 (\lambda-5) (\lambda-7)^3 = 0$
the No. of EV? (MSQ)

a) At least 6

b) At most 6

c) exactly 6

d) At least 3

→ Atleast 3 will be there and maximum can be 6.

So. if $(\lambda-x_1)^{m_1} (\lambda-x_2)^{m_2} \dots (\lambda-x_k)^{m_k}$

then Atleast k LI EVs

Atmost $\sum_{i=1}^k m_i$ LI EVs

Ex 1: Find EV for I.

So. $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)^3 = 0$$

$\lambda = 1, 1, 1$ (Max-3, Min-1)

So. SOL

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad x = \begin{bmatrix} s \\ t \\ u \end{bmatrix}$$

$$x = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

So. 3 LI EVs.

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Ex:2 $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ $\lambda = 1, 1, 1$

then $x = \begin{bmatrix} s \\ t \\ -t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$

∴ Two LI E vectors

Ex:3 $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ $\lambda = 1, 1, 1$

then $x = \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix} = k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

∴ One LI vector

So, Ex.1, Ex.2 & Ex.3, $\lambda = 1, 1, 1$, but no. of LI EV are different.

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Que: Let's have Matrix $A_{4 \times 20}$. We solve eq. $(A - \lambda I)X = 0$ for $\lambda = 3$ so we found 7 free var. then for $\lambda = 3$, how many LF vectors?

$\Rightarrow$ 7 (just Rev. of last que)

Ans: Above que, what could be power of $(\lambda - 3)$ in the characteristic equation (MSQ)

g) 3 h) 7 i) 5 j) 20
 $\Rightarrow$ $20 \leq \text{power} \leq 7$

★ In $(\lambda - x)^m$,

- $\rightarrow$ Arithmetic Multiplicity = m (like 3 in Ex 1, 2, 3)
 $\rightarrow$ Geometric Multiplicity = no. of LF EV. (at most m)
 (3 in Ex 1, 2 in Ex 2 & 1 in Ex 3)

So $AM \geq GM$

AM & GM are properties of E Values.

★ What is the condition under which we will have n independent E vectors even with repeating Eigen values. In other words, Are there matrices which will always have n LF EVs even when Eigen values are repeating? (i.e. $AM = GM$)

$\Rightarrow$ Yes, Symmetric Matrices (of size $n \times n$)

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So, Real Symmetric Matrices have following two properties:

- 'n' real eigen values
- 'n' orthogonal (means parallel) Eigen Vectors
(Parallel EVs are always LE)

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So, for real symmetric matrix, all Eigen Values are real and all eigenvectors are linearly independent (even if eigen values are repeating)

Que: For Real Sym. Matrix $(d-1)^3 (d-5)^7 (d-11)^1$, then, No. of LE EVs?
⇒ Exactly 11.

Que: If, number of free variables in $(A - \lambda_1 I)x = 0$ are 2, then what is the GM of λ_1 ?
⇒ 2

Que: In above question AM of λ_1 is?
⇒ Atleast 2.

Que: Let A be a 5×5 matrix and rank of $A - \lambda_1 I$ is 2, then GM of λ_1 ?

⇒ As rank = 2
free var = $5 - 2$
 $= 3$

So GM = 3

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T/F?

Ques: A matrix of size 5×5 has only 2 LF EVs then it must have repeated eigen values.

$\Rightarrow$ Yes, True, Possible char. eq. are

$\rightarrow (d-d_1)^5$	$\rightarrow (d-d_1)^3 (d-d_2)^2$
$\rightarrow (d-d_1)^4 (d-d_2)^1$	$\rightarrow (d-d_1)^1 (d-d_2)^4$
$\rightarrow (d-d_1)^2 (d-d_2)^3$	$\rightarrow (d-d_2)^5$

Imp.
★
///

Two important & magical (not really) properties:

- Determinants of any product matrix is product of eigen values.
- Trace (sum of elements in main diagonal) of matrix is sum of eigen values.

So $|A| = d_1 d_2 \dots d_n$

$\text{trace}(A) = d_1 + d_2 + \dots + d_n$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$|A| = ad - bc$

$\text{trace}(A) = a + d$

Eigen Values:

$$\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} = 0$$

$\Rightarrow (a-\lambda)(d-\lambda) - bc = 0$

$\Rightarrow \lambda^2 - \lambda(a+d) + (ad-bc) = 0 \quad \text{①}$

As we know in $ax^2 + bx + c = 0$

$$\text{Sum of roots} = x_1 + x_2 = -b/a$$

$$\text{Product of roots} = x_1 \cdot x_2 = c/a$$

Using this in (1)

$$\lambda_1 + \lambda_2 = a + d = \text{trace}(A)$$

$$\lambda_1 \lambda_2 = ad - bc = |A|$$

Hence Properties Proved.

Ques:

$$\text{If } A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

& Sum of E value is m
ie product of E value is n
then m+n is equal to:

$$\Rightarrow \text{As product of E values} = \det(A) = n = 5$$

$$\text{ie Sum of E value} = \text{trace}(A) = m = 7$$

$$\text{So } m+n = 12$$

Ans

Ques: Let A be an nxn matrix with char. polynomial

$$p(d) = d(d-2)(d-3)^2$$

Which of the following can you determine from this info? (Msa)

✓ Number of n

→ Yes, $(1+1+2) = 4$

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Q3. The trace of A

→ Yes, $(0+2+3+3 = 8)$

Q4. The determinant of A

→ Yes $(0 \times 2 \times 3 \times 3 = 0)$

Q5. Whether A is symmetric

→ No.

Q6. Eigen values of A

→ Yes $(0, 2, 3, 3)$ Ques:One of the eigen values of $A = \begin{bmatrix} 3 & 1 & 4 \\ c & 1 & 5 \\ c & 1 & 5 \end{bmatrix}$ is $\lambda = 3$, c is some constant.

Find other two E. values.

⇒ As R_2 & R_3 are same, $\det(A) = 0$ so for that $\lambda_2 = 0$ & also.

$$\lambda_1 + \lambda_2 + \lambda_3 = \text{trace}(A)$$

$$3 + 0 + \lambda_3 = 9$$

$$\lambda_3 = 6$$

So

$$\underline{6 \text{ \& } 0}$$

Ques:TIF If columns of matrix are LD, then one of the E values has to be zero.

→

True, bcz, two LD cols of matrix make determinant 0, which means one of E value should be zero.

Rank & Eigen Values

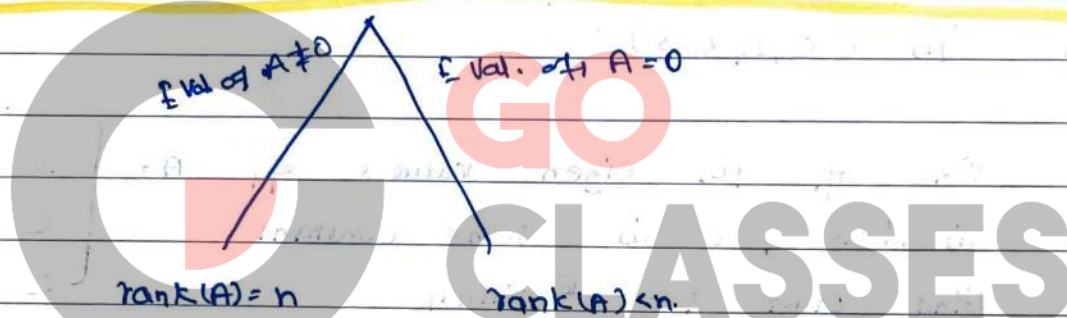
may be complicated
see lecture

★ Relationship of Rank with Eigen Values: Only relation is this:

If ^{each} eigenvalue of $A \neq 0$, then $\text{Rank}(A) = n$.

If λ eigenvalue of $A = 0$, then $\text{Rank}(A) < n$

↳ Bcz, if $\text{EVal} = 0$, then $\det = 0$ ^(not to) for $\det = 0$, we need atleast one free var. $\therefore$ so $\text{rank}(A) < n$.
but not sure if it will be $n-1, n-2$, etc.
but it will be $< n$.



↳ No. of LF EV correspond to $\text{EVal} = 0 = k$
Free val $= k$
 $\text{Rank} = n - k$

★ If $Ax=0$, then can $Ax=0$ have a unique non-trivial soln?

⇒ No, if one soln is present then many other LD soln of that will be there.
So either ~~no~~ zero Non Trivial soln or infinite Non trivial soln.

(Basic, Do not Panic)

Cases: If A is LD,
then 0 non-trivial soln

Case 2: If A is not LD
then ∞ soln.

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★ In $(A - \lambda I)x = 0$, then irrespective of value of λ , there is always atleast one free variable bcz, $(A - \lambda I)$ needs to be zero, for that one of the EV need to be zero i.e. so rank $< n$ i.e. three will be free var.

✖ $(A - \lambda I)x = 0$ & $Ax = 0$ are both diff. SOLS. They will be same if and only if $\lambda = 0$

Ques. There are two free var in $Ax = 0$ soln. then rank of A if A is 10×10

$$\Rightarrow 10 - 2 = 8$$

Ques. There are two free var. when solving $(A - \lambda I)x = 0$ corresponding to $\lambda = 0$, then rank of A if A is 10×10

$\Rightarrow$ if $\lambda = 0$, then $(A - \lambda I)x = 0$
then $Ax = 0$

then same as above que = 8

Ques. If $\lambda = 0$ i.e. No. of LI eigen vectors for $\lambda = 0$ is 2 then rank?

$\Rightarrow$ if LI EV = 2
then free var = 2.
So rank = 8

$$(A - \lambda I)x = 0$$

$$\hookrightarrow Ax = 0$$

$$\hookrightarrow n = 10 \begin{bmatrix} \end{bmatrix} + 2 \begin{bmatrix} \end{bmatrix}$$

$$\hookrightarrow \text{Rank} = 10 - 2 = 8$$

No. of LF EV corresponding to $\lambda=0 \Rightarrow$ Nullity.

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Ques: Let A be 5×5 matrix and one of the Eigenvalue of A is 0. It is also known that there are 4 LF EV corresponding 0 Eigenvalue. What is rank of A ?

Ans: As for $(A - \lambda I)x = 0$
 $\lambda = 0$,
 $Ax = 0$,
 then we have 4 LF soln for x ,
 So free var = 4.
 $\text{rank} = \text{Dim} - \text{free var.}$
 $= 5 - 4$
 $= 1$
 $\underline{\underline{= \text{Ans.}}}$ ✓

Ques: Let A be a 5×5 matrix as it has 4 LF EV. What is rank of A ?

$\Rightarrow$ Can't Comment any thing bcz $(A - \lambda I)x = 0$ & $Ax = 0$ are diff. So L.E. ✓

Ques: Let A be 15×15 matrix & one of its eigenvalue is zero. What is rank of A ?

$\Rightarrow$ Less than 15. ✓

Ques: Let A be 15×15 matrix & one of its Eigenvalue is zero. It is also known that there are 10 LF EV of A . What is rank of A ?

$\Rightarrow$ We don't have idea of all Eigenvalues. We may have more Eigenvalue and may have more LF EV of A . So can't say until all Eigenvalues are told. (~~$\text{Rank}(A) \geq 10$~~)

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Ques.

⇒

Can't say bcz, we have 10 LI EV. But, we don't know how many of the EV are corresponding to Eigenvalue zero.

Ques.

Let A be 15×15 matrix & one of its Eigenvalue is 0. It is also known that there are 10 LI EV of A & out of which 4 LI vectors corresponds to 0 eigenvalue. What is rank of A .

⇒

$$15 - 4 = 11$$

Ques.

(MCQ)

Let A be an $n \times n$ matrix with char eq. $-d(d-1)^2$. Explain whether or not the following can be true, & if it can, given an example.

a) Rank $(A) = 0$

b) Rank $(A) = 1$

c) Rank $(A) = 2$

d) Rank $(A) = 3$

⇒

$$\text{Here } n = 1 + 2$$

$$= 3$$

As $A \neq I$ as in char eq. $-d$ has power 1 then exactly one will be the LI EV for $d=0$

$$\text{So Rank}(A) = 3 - 1$$

$$= 2$$

Ques.

In above que. if char eq. $d^2(d-1)^3$, then Rank (A) could be?

⇒

$$n = 5, \text{ any free var} = 1, 2 \text{ (AMR} = 2)$$

$$\text{Rank} = 5 - 1, 5 - 2,$$

$$= 3, 4$$

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Ques: In prev. que if matrix was sym. then Rank was?

$$\Rightarrow n=5,$$

$$AM=2,$$

$$\text{ie } GH=AM=2$$

($\therefore$ Sym Matrix)

$$\text{Rank} = \underline{\underline{3}}$$

100
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(CHT)

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation.

i.e. if $\phi(\lambda) = \lambda^2 - \lambda + 5 = 0$ for matrix A then $A^2 - A + 5I = 0$ is also true.

Ques:

Suppose the char. eq. of A is $f(t) = t^4 - 1$.
Use CHT to express $A^{10} + A^8$ as a LC of I, A, A^2, A^3 .

$$\Rightarrow \text{As } t^4 - 1 = 0$$

$$\text{So } A^4 - I = 0$$

$$A^4 = I$$

Sq. both sides

$$A^8 = I$$

multiply by A^2

$$A^{10} = A^2$$

So

$$\underline{\underline{A^{10} + A^8 = A^2 + I}}$$

$\therefore$ Expressed.

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Ques: The char. poly. $p(\lambda)$ of a 2×2 matrix B is
 $p(\lambda) = \lambda^2 - 3\lambda$

Using CHT, we can say that:

a) $B^n = 3^{n-1} B$

b) $B^n = 3^{n+1} B$

c) $B^n = 3^n B$

d) $B^n = B^{3^n}$

e) NOTA

$\Rightarrow$ As $B^2 - 3B = 0$ (From char. poly.)

$B^2 = 3B$ — (1)

$B^3 = 3B^2$ (Multiply by B)

$B^3 = 3(3B)$ (From eq. (1))

$B^3 = 3^2 B$

So

$B^n = 3^{n-1} B$ — Option (a)

★

Uses of CHT:

$\rightarrow$ To calc. the positive integral powers of A .

$\rightarrow$ To calc. the inverse of a square matrix A .

Ques: $A = \begin{bmatrix} 3 & 0 \\ 4 & 1 \end{bmatrix}$

a) Use CHT to find A^{-1} .

$\Rightarrow \begin{vmatrix} 3\lambda & 0 \\ 4 & 1-\lambda \end{vmatrix}$

$\Rightarrow (3-\lambda)(1-\lambda) = 0$

$3 - \lambda + \lambda^2 - 3\lambda = 0$

$\lambda^2 - 4\lambda + 3 = 0$ — char eq.

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So $A^2 - 4A + 3I = 0$ - (1)

$$A(A - 4I) + 3I = 0$$

So $A(A - 4I) = -3I$

$$A \cdot \frac{(A - 4I)}{-3} = I$$

(You need to come to this form) $(A \cdot X = I)$
then $A^{-1} = X$

So $A^{-1} = \frac{A - 4I}{-3}$

$$= -\frac{1}{3} \left\{ \begin{bmatrix} 3 & 0 \\ 4 & 1 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$$= -\frac{1}{3} \begin{bmatrix} -1 & 0 \\ 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 & 0 \\ -4/3 & 1 \end{bmatrix}$$

(Confirm by $AA^{-1} = I$)

b) Use CHT to find A^2

$\Rightarrow$ As $A^2 - 4A + 3I = 0$

$$A^2 = 4A - 3I$$

$$= 4 \times \begin{bmatrix} 3 & 0 \\ 4 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 \\ 16 & 1 \end{bmatrix}$$

Ques.

Let A be a 3×3 matrix. If A has E values $0, 0, 0$ then $A^3 = 0$.

$\Rightarrow$ for E values $0, 0, 0$,

Char Eq. $\Rightarrow (A - 0)^3 = 0$

$$= A^3 = 0$$

& By CHT

$$A^3 = 0$$

True

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Que. Let A be a 2×2 Matrix. Char. Eq. is $\lambda^2 - 2\lambda + 5$. Use CHT to find a & b so that $A^3 = aA + bI$.

$\Rightarrow$ By CHT

$$\lambda^2 - 2\lambda + 5 = 0$$

$$\lambda^2 = 2\lambda - 5 \quad - (1)$$

$$\lambda^3 = 2\lambda^2 - 5\lambda \quad (0 \leq \lambda)$$

$$\lambda^3 = 2(2\lambda - 5) - 5\lambda \quad (\text{Use } (1))$$

$$\lambda^3 = 4\lambda - 10 - 5\lambda$$

$$\lambda^3 = -\lambda - 10$$

So, $a = -1$ & $b = -10$

Que. The Char. eq. is $\lambda^2 - 5\lambda + 5 = 0$.

NSQ.

Q. What does the CHT imply about the exp. $B^2 - 5B + 5I$?

1) It equals I

2) It is composed of EV of B

3) It is undefined

4) It equals B

5) It equals 0

$\Rightarrow$ 5 only

b) We use CHT to compute $2B^3 - 10B^2 + 10B - 5I$

$\Rightarrow$ We have

$$B^2 - 5B + 5I = 0$$

$$B^2 = 5B - 5I$$

$$B^2 = 5B - 5I \quad - (1)$$

$$B^3 = 5B^2 - 5B \quad - (B \times (1))$$

$$B^3 = 5(5B - 5I) - 5B = 25B - 25I - 5B = 20B - 25I$$

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$$2B^3 = 10B^2 - 10B \quad (10 \times 2)$$

$$2B^3 - 10B^2 + 10B = 0$$

$$2B^3 - 10B^2 + 10B - 2I = -2I \quad (\text{Subtract } 2I)$$

$$\text{So ans} = \underline{\underline{-2I}}$$

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★

Eigen Values of AB & BA: Let λ be nonzero
Eval of AB & x be EV
of AB. What will be the eval & EV of BA?

$$\Rightarrow ABx = \lambda x \quad (\text{Given}) \quad \text{--- (1)}$$

$$BABx = \lambda Bx \quad (\text{Premultiply B both sides})$$

How we can infer from above eq. that
 Bx is eigen vector (EV) of BA with λ as eval.

Also we need to ensure that Bx is
non zero before inferring above line. But this
cannot be zero (see eq- (1)) as if Bx is
zero then λx will be zero, but it
can't be zero.

Remark

So. ★ AB & BA share non zero E values.

Now let $A_{n \times n}$ & $B_{n \times n}$ be two matrices and
Evals of $AB_{n \times n}$ are $\lambda, \mu, \gamma, \delta, \dots$ then the
Evals of $BA_{n \times n}$ are $\lambda, \mu, \gamma, \delta, \dots$ & 0.

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Ques. $A_{2 \times 10} \in B_{10 \times 2}$, Eval. of $AB_{2 \times 2} \Rightarrow 1, 1, 5$
 then Eval. of $BA_{10 \times 10} \Rightarrow 1, 5, \underbrace{0, 0, 0, 0, 0, 0, 0, 0, 0, 0}_6$

there can be zero only
 bcz if they were nonzero, they would also been
 in $AB_{2 \times 2}$.

Ques. $A_{4 \times 3} \in B_{3 \times 4}$, (Eval. of $AB_{4 \times 4} \Rightarrow 1, 2, 3, 4$
 atleast.
 then this que. is wrong, One Eval. of AB will
 be zero.

Ques. $A_{4 \times 3} \in B_{3 \times 4}$, Eval. of $AB_{4 \times 4} \Rightarrow 1, 2, 0, 0$
 then Eval. of $BA_{3 \times 3} \Rightarrow 1, 2, 0$

Ques. $A_{4 \times 3} \in B_{3 \times 4}$, then prove that AB must have
 atleast one zero eval.

$\Rightarrow$ D.I.Y.

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Ques. Consider the following 2 statements w.r.t. matrix
 $A_{m \times n}$, $B_{n \times m}$, $C_{n \times n} \in D_{n \times n}$.

S1: $\text{tr}(AB) = \text{tr}(BA)$

S2: $\text{tr}(CD) = \text{tr}(DC)$

Where $\text{tr}()$ is trace. Which one holds?

a) S1 $\checkmark$ S2 $\times$

b) S1 $\times$ S2 $\checkmark$

c) S1 $\checkmark$ S2 $\checkmark$

d) S1 $\times$ S2 $\times$

$\Rightarrow$ Trace is sum of eval. & it will not be
 affected. as adding zero don't matter.

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Que.

Consider a matrix $A = UV^T$ where $U = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $V = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Largest eval of A is _____?

$\Rightarrow$ M1: $UV^T = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$

d of $UV^T = \begin{bmatrix} 1-d & 2 \\ 1 & 2-d \end{bmatrix}$

$(1-d)(2-d) = 0 \Rightarrow d(d-3) = 0$

$d=0$ or $d=3$, So Answer is 3

Efficient / FastM2:

Instead of UV^T find EV of UV^T or BA or AB
Share same Eval:

$UV^T = [3]_{1 \times 1}$

$(3-d) = 0$

$d=3$

Even if U was 1×1000 or V was 1×1000 , this could be solved easily by M2.

★ 100.

Eigen values of Powers of A: If Eval of A is d , then what will be the Eval of A^k ?

$Av = \lambda v$

— (1) (Given)

$A^2v = \lambda Av$

(Multiply (1) with A)

$A^2v = \lambda^2 v$

(As $Av = \lambda v$)

Sim. $A^3v = \lambda^3 v$

$A^3v = \lambda^3 v$

So

$A^k v = \lambda^k v$

Ans

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So. If Eval of A is d , then Eval of A^k will be d^k as it will be EU for both.

Que. If Eval of A is d , then Eval of $A^k + 3A^{k-1} + I$ is?

$$\begin{aligned} \Rightarrow (A^k + 3A^{k-1} + I)x &= A^k x + 3A^{k-1} x + x \\ &= d^k x + 3d^{k-1} x + x \\ &= x(d^k + 3d^{k-1} + 1) \end{aligned}$$

So Eval is $d^k + 3d^{k-1} + 1$

So we can directly write d in place of A .

Que. If Eval of A is d , then Eval of A^{-1} is?

$\Rightarrow$ Directly d^{-1} or $1/d$ or

$$Ax = d x \quad \text{--- (1)}$$

$$x = d A^{-1} x \quad \text{(1) } \times A^{-1}$$

$$\frac{1}{d} x = A^{-1} x$$

$$A^{-1} x = \frac{1}{d} x \quad \text{So } 1/d \text{ is eval.}$$

* If A is a $n \times n$ matrix & $(d_1, d_2, \dots, d_k)$ are its evals, then the evals of $A + I$ are $(d_1 + 1, d_2 + 1, \dots, d_k + 1)$ & for polynomial P , the eval of matrix $P(A)$ are $\{P(d_1), \dots, P(d_k)\}$.

— Wikipedia.

Vector $\begin{bmatrix} x \\ y \end{bmatrix}$ & $\begin{bmatrix} -y \\ x \end{bmatrix}$ are perpendicular to each other or orthogonal

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Que. for $SA^3 + A^2 - A + I$

$$\Rightarrow 5A^3 + A^2 - I$$

Que. Do we always have eigen values & eigen vectors?

$\Rightarrow$ No. Ex: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$

$\hookrightarrow$ This is rotating every vector
so, there will not be any EV
for this.

Proof:

$$\begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i \text{ (iota)}$$

So λ is imaginary.

Also, here it comes in pair, i.e. i & $-i$.

NOTE $\longleftrightarrow$

Let A be an $n \times n$ matrix with real coeff.
If λ is eval of A & V as Ass. EV, then
 $\bar{\lambda}$ is also an eval with Ass. EV as $\bar{V}$.
(V & $\bar{V}$ are conjugates of V & λ)

So, to be proved: $AV = \lambda V \Rightarrow A\bar{V} = \bar{\lambda} \bar{V}$

$$AV = \lambda V \quad (\text{Given})$$

$$\bar{A}\bar{V} = \bar{\lambda}\bar{V} \quad (\text{take conj. both sides})$$

$$A\bar{V} = \bar{\lambda}\bar{V} \quad (\text{As } A \text{ is real matrix})$$

L → Lower Half
U → Upper Half

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LU Decomposition: Method to decompose matrix A into L & U.

$$A_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ * & 1 & 0 \\ * & * & 1 \end{bmatrix} \quad \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

L U

* → Non zero

★ LU Decomposition is only possible if A is able to be reduced to row echelon form, U, without interchanging any rows.

→ Two Methods for LU Decomp:

★ Short Method:

$$A = \begin{bmatrix} 1 & 4 & -3 \\ -2 & 8 & 5 \\ 3 & 4 & 7 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ * & 1 & 0 \\ * & * & 1 \end{bmatrix}$$

Step 1 Convert A to Echelon Form.

(Also cannot swap rows & op.

should be of form $R_i \rightarrow R_i + kR_j$, i.e.

$R_i \rightarrow \lambda R_i + kR_j$ is not allowed

(& update -k in L)

$$R_2 \rightarrow R_2 + 2R_1 \rightarrow \begin{bmatrix} 1 & 4 & -3 \\ 0 & 16 & -1 \\ 3 & 4 & 7 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ * & * & 1 \end{bmatrix}$$

Proof of short method is long & not req. so not told.

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$$R_3 \rightarrow R_3 - 3R_1 \rightarrow \begin{bmatrix} 1 & 4 & -3 \\ 0 & 16 & -1 \\ 0 & -8 & 16 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & * & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + \frac{1}{2}R_2 \rightarrow \begin{bmatrix} 1 & 4 & -3 \\ 0 & 16 & -1 \\ 0 & 0 & 3\frac{1}{2} \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -\frac{1}{2} & 1 \end{bmatrix}$$

↳ This is U ↳ This is L.

L is made in side when A is converted to echelon form.

(Also can verify by mul. U & V.)

★ Long Method:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Multiply L & U

$$\begin{bmatrix} 1 & 4 & -3 \\ -2 & 8 & 5 \\ 3 & 4 & 7 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

↳ then get u_{ij} to v_{ij} .

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★ Use long only if you forget short & we can derive long not short

Ques: Find LU decomp. of $A = \begin{bmatrix} 1 & 0 & 1 \\ a & a & a \\ b & b & a \end{bmatrix}$ when it exist

$$\xrightarrow{R_2 - aR_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ b & b & a \end{bmatrix} \quad \checkmark \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{b}{a} & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 - bR_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ 0 & b & a-b \end{bmatrix} \xrightarrow{R_3 - \frac{b}{a}R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ 0 & 0 & a-b \end{bmatrix}$$

So $a \neq 0$

Δ → Always Verify also

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For any questions or issues regarding my handwritten notes, feel free to reach out via email at karan757527@gmail.com or DM me on telegram at @karan757527 (or by clicking [here](#)).

I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click [here](#) to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal